Evaluating the Effectiveness of Visual Representations of SHAP Values Toward Explainable Artificial Intelligence

Bianca Moreira Cunha
Departamento de Informática
Pontifícia Universidade Católica do Rio de Janeiro
Rio de Janeiro, Rio de Janeiro, Brasil
bcunha@inf.puc-rio.br

Simone Diniz Junqueira Barbosa*
Departamento de Informática
Pontifícia Universidade Católica do Rio de Janeiro
Rio de Janeiro, Rio de Janeiro, Brasil
simone@inf.puc-rio.br

Abstract

The employment of Machine Learning (ML) and Deep Learning models across diverse domains has grown exponentially in recent years. These models undertake critical tasks spanning medical diagnoses, criminal sentencing, and loan approvals. Consequently, there is a need for these models to be interpretable, enabling users to grasp the rationale behind predictions and engendering trust. Equally vital is the capability of developers to pinpoint and rectify any erroneous behaviors. In this context emerges the field of Explainable AI (XAI), which aims to develop methods to make ML models more interpretable while maintaining their performance level. Various methods have been proposed, many leveraging visual explanations to elucidate model behavior. However, a notable gap remains: a lack of rigorous assessment regarding the effectiveness of these explanations in enhancing interpretability. In this paper, we evaluate whether the SHAP method, widely used in the XAI field, achieves its objective of making a model more interpretable. We conducted a study based on the concept of *simulatability*. We asked participants who have mathematical or statistical backgrounds and intermediate to advanced knowledge of machine learning and model interpretability to classify some given instances of a dataset, presented first without the explanations provided by the SHAP method and then with them. The goal was to assess whether the explanation made the participants better at classifying the instances. Our findings reveal that the visualizations can be confusing even for users with a mathematical background. Additionally, we argue that there is a need for XAI researchers to work collaboratively with visual analytics experts to develop these visualizations, as well as test the visualizations with users of various backgrounds.

CCS Concepts

• **Human-centered computing** → **Visualization**; **Visualization techniques**; **Empirical studies in visualization**.

Keywords

Visualization, Explainable Artificial Intelligence, SHAP values

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
IHC 2024, October 07–11, 2024, Brasília - DF, Brazil
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1224-1/24/10
https://doi.org/10.1145/3702038.3702093

ACM Reference Format:
Bianca Moreira Cunha and Simone Diniz Junqueira Barbosa. 2024. Evaluating the Effectiveness of Visual Representations of SHAP Values Toward Explainable Artificial Intelligence. In *XXIII Brazilian Symposium on Human Factors in Computing Systems (IHC 2024), October 07–11, 2024, Brasília - DF, Brazil*. ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3702038.3702093

1 Introduction

Systems that use Artificial Intelligence (AI) and Machine Learning (ML) have been quickly becoming increasingly available and adopted in various fields in the past years. Many of the tasks they intend to support are tasks of high risk or crucial importance for people, companies, or society as a whole. For instance, systems used in the medical, legal, or security fields are ones in which the user demands a high level of trust in the system and needs to be able to understand why a certain decision was made by the model and to know when it is not behaving as expected. To gain that trust, it is not sufficient to provide accurate predictions [17]; trust requires an understanding of the model behavior [12]. Explanation methods such as SHAP [19] and LIME [25] have been proposed to generate explanations for the behavior of a model or for specific predictions that will help the user to understand how the model reaches its decisions.

Many of these methods leverage the power of data visualizations to communicate valuable insights to convey the information provided by them. Graphical representations for the explanations generated by these methods have been shown to be useful for improving users' trust in ML solutions [23]. However, many of these methods claim to improve the models' interpretability without actually evaluating whether that assertion is true. Furthermore, it has not been long since researchers have proposed a formalization for the meaning of interpretability, and there are few works that propose a way to assess the interpretability of a model or an explanation generated by the previously cited methods.

Doshi-Velez and Kim [7] provide a definition for interpretability and propose a taxonomy of interpretability evaluation. Based on the concept of *simulatability* proposed by them, Hase and Bansal [10] developed an experiment that assesses how the users' understanding of a model is affected when they have access to explanations generated by the methods LIME and Anchor [25, 26].

In this paper, we conducted an experiment inspired by Hase and Bansal [10], intending to assess how the visualizations of the explanations generated by the SHAP method [19] affect people's understanding of the model. We take a step further and include a textual explanation of the visualization, with the goal of evaluating

the differences in understanding between having only a visualization, which can be hard to understand for people that are not familiar with data charts and having an additional verbal explanation. e aim to contribute with evidence that proves if the existing visualizations for explanations generated by one of the most used explanation methods is efficient in achieving it's goal or not, and therefore justify that the assessment of such explanations is of utmost importance.

The remainder of this paper is organized as follows. Section 2 presents the concepts of and foundations used in the paper, as well as related work on how visualization may contribute to the understanding of AI models. Section 3 describes the study we conducted. Section 4 discusses the results of the study and Section 5 describes some limitations of the study and how we suggest to overcome them in future studies. Finally, section 6 concludes the paper with an appraisal of our contributions and future research directions.

2 Background and Related Work

2.1 What does interpretability mean?

To assess the interpretability of a model or whether a certain method can make a model more interpretable, we need first to understand what *interpretability* means. Doshi-Velez and Kim [7] and Lipton [17] came to the conclusion that there is no consensus in the literature as to the meaning of interpretability or how to evaluate it. Therefore, the claim that a model is interpretable has a quasi-scientific character [17]. Doshi-Velez and Kim [7] proposed a definition for the term and a taxonomy of interpretability evaluation. Other researchers have proposed a few meanings for interpretability in the context of machine learning.

Kim [12] stated that interpretable models enable communication from machines to humans without changing the internal states of models. Hase and Bansal [10] suggested that a model is interpretable if it is simulatable. Chatzimparmpas et al. [5], Kim [12], and Ridgeway et al. [27] asserted that interpretability is a means to engender trust. Doshi-Velez and Kim [7] said that interpretability can be defined as the ability to explain something or to present it in understandable terms to a human. Biran and Cotton [4], Mittelstadt et al. [22], and Lou et al. [18] understand an interpretable model to be one of which a person can have an understanding of its inner mechanisms. The latter also proposes that a model can be interpretable if a person can understand a specific prediction, even if they do not understand how the model functions. These two ways of understanding interpretability proposed by Lou et al. [18] derived the concepts of *ad-hoc* and *post-hoc* interpretability, and they will be further explored in subsection 2.3.1.

Kim et al. [13] gave a formal definition to "interpretation" of a ML model as a function $g : E_m \rightarrow E_h$, where E_m and E_h are vector spaces. The first is composed of basis vectors e_m corresponding to data such as input features and neural activations, and the second is composed of basis vectors e_h corresponding to an unknown set of human-interpretable concepts.

From these definitions, it is possible to reason that interpretability in the context of machine learning refers to being able to trust a certain model because you can understand how it generated a specific output.

2.1.1 Interpretability vs Explainability. In the literature, we could encounter two concepts that are similar and are often used interchangeably but that have a subtle difference in meaning: "interpretability" and "explainability." Lipton [17] considers interpretability as a measure of transparency of a model,*i.e.*, the understanding of how the model works internally, how it gets to certain decisions. Transparency can be viewed as the opposite of opacity or black box-ness. In contrast, he defines post-hoc explanations as further information that we can get from a model, such as the significance of various parameters. Miller [21] uses the definition given by Lipton [17] and considers explanations as a means by which an observer may obtain an understanding of a model. In this paper, we understand interpretable models as models in which an observer can understand their inner workings, *e.g.*, a simple linear model with few coefficients can be interpretable since it can be easily understood. As for explainability, we consider an explanation to be an instrument that assists the observer in the understanding of a model by giving them additional information. That is what Explainable AI (XAI) aims to do: to propose methods that will provide explanations for models that are not naturally interpretable. This is addressed in the next section.

2.2 Why is interpretability desired?

Having the proposed definition of interpretability in mind, we need to understand why there is a necessity for interpretable models. Why could we not settle for accurate models, knowing that we can measure their correctness, even if we do not understand why they reached a specific result? The desire for model interpretations indicates that a model's predictions and its accuracy metrics are not sufficient to characterize the model [17].

Machine Learning models are applied in many high-stakes scenarios, so the need for interpretable models will only grow. In these situations, domain experts need to be able to understand and trust the model they are working with so that they can make important decisions [15]. Some real-world problems are hard to formalize, and that is when interpretations are needed [17]. Hoffman et al. [11] argued that "XAI systems should enable the user to know whether, when, and why to trust and rely upon the XAI system and know whether, when, or why to mistrust the XAI and either not rely upon it, or rely on it with caution."

Another reason brought up by Tamagnini et al. [31] for the need for model interpretability is that a model that analysts can inspect and observe its decisions can support them in understanding better the data and the phenomenon it describes. Furthermore, understanding the predictions given by a model can allow the user to identify possible causes of errors, misclassifications, or points of unfairness [6, 23].

Doshi-Velez and Kim [7] argued that interpretability is a way to qualitatively assess whether objectives such as fairness, privacy, reliability, robustness, causality, usability, and trust are met. They also debate that the need for interpretability comes from an incompleteness in the problem formalization, which causes an obstacle to optimization and evaluation. Therefore, explanations are a way to make gaps in problem formalization visible so they can be addressed.

Apart from the previously cited practical reasons, there are also strategic motivations for having explanations for AI systems' decisions. There is a higher propensity for users to trust and rely on a system if they can understand its outputs, and therefore, a higher probability for them to adopt the system [32]. Additionally, the European Union law for data protection and privacy, the General Data Protection Regulation, released in 2016, states that AI systems must provide "meaningful information about the logic involved" in the decision-making process and also should provide "an explanation of the decision reached" [24]. Correll [6] also brings up the moral duty in legal cases to inform the impacted people of the decision-making process.

2.3 Explainable AI

Simple models, such as linear regression or simple tree models, can be more interpretable, but they can also be less accurate than more complex models [19]. Some argue that even these less complex models cannot be interpretable when they are sufficiently high-dimensional [17]. Interpretability is desired; however there is also a need for model performance. Therefore, several methods have been proposed in the past years with the purpose of generating explanations for machine learning models' behavior or for the outputs they produce and thus improving the models' interpretability without having performance loss. The field of AI encompassing this kind of work in literature is called Explainable AI (XAI) [2]. The Defence Advanced Research Project Agency (DARPA) technical report [8] defined XAI as "a suite of machine learning techniques that enables human users to understand, appropriately trust, and effectively manage the emerging generation of artificially intelligent partners". The focus of the field is on improving the transparency of how the models get to their outputs, aiming to provide clarity to computer programmers and expert users [3, 32].

A characteristic of current AI explanations presented by Woodcock et al. [32] is that they give answers to *how* questions and not to *why* questions, even though the last ones are people's preferences. These methods will then answer the question "*How* did you get to that decision?" The cited work presents two ways in which the *how* questions are answered:

- **Input influence**, which presents the list of input variables with an impact value for each of them. It is also called a local explanation [10, 19, 26, 31].
- **Case-base reasoning**, which displays an instance of the training set similar to the one being classified [10].

Both methods require the human recipient to have prior domain knowledge so that they can understand and evaluate the explanation. The method of input influence also has the downside that long lists of input feature contributions might be hard to interpret [6].

Woodcock et al. [32] stated that *why* questions are preferred by humans, and they are implicitly contrastive, which means that the answer to them offers an explanation relative to some event that did not happen. They call that event a *foil*. They consider the core finding of their work to be that "in order to close a user's information gap, the AI explanation must be generated with an understanding of that user's unique foil."

Explanations have been explored only recently in the AI field but have been largely studied in other fields. These studies should be considered when developing explainability methods [11]. An important factor to be considered in Explainable AI is curiosity since it is one of the core reasons for people to seek explanations. Therefore XAI systems must take advantage of the power of curiosity [11]. Furthermore, Kim [12] raised the concern that the traditional machine learning systems are not designed with the possibility to leverage the knowledge of domain experts with their direct interaction with the system. In order for these systems to make real impact in important domains, machine learning and humans should take advantage of their complementary skills and work collaboratively.

Li et al. [16] created a visual analytics system that helps users compare ML models in terms of their performance, consistency, and reliability. They present the system by utilizing a medical dataset as a case study and explain how their visualizations assist the user in understanding how different ML models differ in their prediction process, how each feature impacts each model, and how consistent each of the models is. They claim that they were able to demonstrate the usefulness of their system. However, there is no test with actual end users to support their claim.

2.3.1 Types of interpretability methods. There are two ways of working with interpretability that have been recognized by literature: ad-hoc and post-hoc interpretation. **Ad-hoc interpretation** explicits a model's inner workings while **post-hoc interpretation** concerns how the model got to a certain prediction [22]. The explanation methods cited before focus mainly on post-hoc explanations. The advantage of post-hoc interpretations is that black-box models can be interpreted without having their predictive performance undermined [17].

Explanations generated by these methods can be textual explanations or visualizations. A language model, for instance, can be trained to generate textual explanations for another model. In contrast, visualizations can be rendered to qualitatively show what a model has learned [17].

Another dimension of explanations divides them into local and global explanations. A **local explanation** will clarify how the model got to a specific prediction, while a **global explanation** explains the model's behavior as a whole. An example of local explanation is an *explanation by example*, which is also a post-hoc mechanism that provides other instances that the model considers to be similar [17].

Post-hoc interpretations, however, present some challenges. First, many models operate on features such as pixel values, which are not easy concepts for humans to understand. Furthermore, the explanations must correctly reflect the model's complex internals [13].

2.3.2 Understanding models through visualization. Visualization has been shown to be an important tool for improving trust in ML solutions, which is not a trivial task [5, 23]. Spinner et al. [30] argued that a natural way of obtaining human interpretable explanations is through visualizations. Graphical explanations help users visualize the evidence for a classification decision efficiently and provide an audit of the model, thus making it possible for the user to identify when a decision is unexpected or erroneous [23]. A variety of approaches have been proposed with the objective of clarifying how certain classes of models function [15].

Chatzimparmpas et al. [5] found that visualization of feature importance, the impact of different characteristics of the data instances, investigation of hyper-parameters, pre-processing steps, and the evaluation of the model are the most popular among users that have the desire to understand the ML process.

Poulin et al. [23] proposed a graphical explanation framework that aims to increase the user's capability to understand and audit the classification process based on evidence. It has five representation capabilities: (1) representing the classification decision; (2) representing the decision evidence, by showing the impact of each feature on each possible decision; (3) decision speculation, where they provide an interactive interface where users can do "what-if" analyses by changing the values of features and see how it would affect the decision; (4) representing ranks of evidence, by ranking the evidence of each feature for the overall behavior of the model; and (5) representing the source of evidence, *i.e.*, the data supporting evidence contributions.

Recent approaches focus not only on visual design but also on interactive, mixed-initiative workflows, as provided by Visual Analytics (VA). The varied backgrounds of the different user groups affected by AI bring various requirements for XAI tools [30]. Domain experts may use XAI tools with the purpose of scrutinizing models and changing hyperparameters to facilitate the AI system's analysis, while AI specialists and developers may use it to discover flaws in the architecture of their models [2].

Alicioglu and Sun [3] named a sub-field of XAI that focuses on VA research, Visual-Based XAI (vXAI). In their review, they found that it is still an under-explored field with a limited number of papers published. They found that local explanations help the user to understand the model behavior for similar instances. Additionally, they discovered that the most popular representations for local explanations are bar charts, breakdown charts (both can be applied in text, tabular, or image data), and heatmaps (mostly for image data). Global explanations are more challenging for black-box models due to their complex structure and computational process. Therefore, there are only a few papers proposing methods that provide global explanations. Histograms are currently the most popular visualization for global explanations. Since vXAI is still a novel field, there is no common visual approach to represent XAI methods for different types of data and models, and there is no standardized way to depict local and global explanations. Researchers tend to develop visualizations customized to their data domain and application area.

2.3.3 Importance of evaluating interpretability. As said before, many explanation methods have been proposed in the past years. However, there has been little analysis of the reliability and robustness of such methods, making their utility for critical applications uncertain. Slack et al. [28] raises some critical questions about these methods that must be repeatedly asked and answered, such as: will the methods make the models more interpretable, trustworthy, and accountable? To whom will the explanations be accessible, comprehensible, or useful?

Current interpretability evaluation approaches rely on some notion of "you'll know it when you see it". There is a need for more rigor in that evaluation since not all models, even in the same class, may be comparable. Likewise, different applications may have different interpretability needs. To evolve the field and to be able to compare explanation methods and to understand when they may generalize, there are demands for a formalization of the notions of what makes a model interpretable and for them to be evidence-based [7]. Hase and Bansal [10] showed some issues we might encounter in explanation methods. First, many explanation methods may not actually help users to understand how a model behaves. Also, a method that works successfully in one domain might not work as well in another domain. Ali et al. [2] agreed with them on that point and also debated that most of the existing approaches are built with explainability aims that are too general and lack well-defined context-specific use cases, and, as a consequence, miss unique needs of a certain domain, resulting in poor adoption and sub-optimal outcomes. Lastly, Hase and Bansal [10] stated that combining information from explanations does not necessarily result in explicit improvements in simulatability.

Spinner et al. [30] argued that the proposed tools are often implemented as standalone prototype solutions and lack integration with the ML developing and debugging process. They claim, therefore, that there is a gap between theory and practice and confirmed in their study that the people developing these XAI methods mostly have no hands-on experience using their proposed tools.

There is also a need to consider the intended audience of the proposed XAI approaches. Most of the existing methods do not state what their intended audience is and, therefore, do not assess if their audience's needs are met [2]. Some of the existing explanation methods generate approximations to the system's behavior. These might be helpful for pedagogical purposes but can also be misleading when presented to lay users [22]. Slack et al. [28] performed some tests that showed that it is possible for malicious actors to build discriminatory models that can fool post-hoc explanation techniques, which shows that these methods can be insufficiently robust for ensuring that the model is trustworthy, reliable, and fair in sensitive applications. A local explanation given to an individual who does not understand the model's limitations can be incomprehensible or even misleading. This kind of explanation can be useful in a specific context and help get to a specific decision, but it does not give insight into how a model functions as a whole. Therefore, Mittelstadt et al. [22] argued that these explanations might be useful for understanding relationships between variables relevant to a particular decision, but they do not prove the model's trustworthiness overall.

2.3.4 Methods to measure and assess interpretability. Ali et al. [2] stated that the value of an explanation is significantly influenced by how valuable it is proven to be for an end user responsible for decision-making. Consequently, there is a need to involve end users in the process of assessing explainability methods, preferably in a context with real tasks and data. They also argued that the user's performance, *e.g.*, accuracy or speed of the decision-making process, should be the measurement for the method's evaluation. Additionally, it is important to consider the user's understanding and satisfaction with the given explanation, and therefore a significant portion of the evaluation relies on qualitative assessments, using surveys or interviews.

Lakkaraju et al. [14] proposed a way to capture interpretability of models based on decision sets. Decision sets are sets of classification rules, where each rule is an independent classifier. They

defined four metrics for measuring interpretability: size, length, cover, and overlap. "Minimizing size encourages decision sets with a small number of rules. Minimizing length captures the notion that interpretable rules are short and concise. We use cover to denote how many datapoints satisfy the itemset of a rule, which is necessary for defining subsequent metrics. Finally, minimizing overlap encourages each rule to cover an independent part of the feature space." They stated that a set of rules is considered interpretable if (1) the rules in the set describe non-overlapping feature spaces; (2) most data points are covered by the rules in the rule set; (3) the set is composed of a small number of rules, and these rules are concise; and (4) the rules in the rule set describe most of the classes present in the data. That is a useful and complete way of assessing interpretability, nevertheless it only encompasses models based on decision sets.

Adebayo et al. [1] also proposed a method to guide researchers in assessing the scope of model explanation methods. Their method is based on randomization tests to evaluate the adequacy of explanation methods and serve as sanity checks in the construction of new explanation methods.

[11] developed a framework to assess explanations by considering the following measurements:

(1) Explanation goodness (They provide a checklist for assessing goodness)
(2) Explanation satisfaction (They provide a scale for assessing explanation satisfaction)
(3) Users' mental models (They provide a list of methods to elicit mental models and ways of assessing them)
(4) Curiosity (They provide a checklist to measure curiosity)
(5) Trust (They give a recommended trust scale for xAI systems)
(6) Performance

Doshi-Velez and Kim [7] proposed a taxonomy of evaluation approaches for interpretability: application-grounded, human-grounded, and functionally-grounded.

- **Application-grounded evaluation** is the conduction of human experiments using real applications, *i.e.*, the explanations are evaluated based on their end-task, such as identifying errors. The method is tested according to the application's objective, thus giving concrete evidence of success.
- **Human-grounded evaluation** involves conducting simpler human-subject experiments while maintaining the nature of the target application. This kind of assessment is useful when doing experiments with the target community is challenging.
- **Functionally-grounded evaluation** requires no human experiment; it considers interpretability as a proxy for explanation quality. This type of experiment is easier to perform since they do not demand the time and cost that human experiments need. They can be used when a method has already been validated by a human-grounded experiment, when a method is not mature, or when human subject experiments are unethical.

For the purposes of this study, we performed a human-grounded evaluation, which will be described in Section 3.

Having the related work in mind, we saw that there is still few studies regarding visualizations for XAI, and we did not find in our research some assessment of what is already available for users in terms of visualizations of explanations generated by the SHAP method. We find it to be a significant evaluation to be done, since this is one of the most known and utilized explanation methods, that already has an implementation available to use in a Python library and there are also online courses that teach people how to use it. Therefore we aim to evaluate the explanations generated by this method with the purpose of generating evidence for it's effectiveness or ineffectiveness and stimulate the assessment of existing and future explanations generated by such methods.

3 Methodology

In this section, we present the study conducted to evaluate whether explanations generated by a known explanation method help people interpret machine learning models or not. We describe the dataset used, respondents pool, and experimental design. This study is part of a research project that was submitted to and approved by the Research Ethics Committee at PUC-Rio, protocol 97/2020. In the online questionnaire, we explained to the participant the goals of the study and its ethical considerations, including the steps we took to respect their privacy and maintain their anonymity in all study reports. We also made clear that they could interrupt or abandon the study at any moment without justification or penalty. The questionnaire included an item for them to declare their informed consent.

3.1 Dataset

The dataset was obtained from the National Institute of Diabetes and Digestive and Kidney Diseases[29]. It has 1,548 records of diagnostic measurements of female patients who are at least 21 years old and of Pima Indian heritage. The target variable is a boolean one that indicates whether the patient was diagnosed with diabetes. With the purpose of not letting the respondents' prior knowledge of diabetes influence their responses, we changed the description of the dataset and said that the data was collected from extraterrestrial beings to diagnose a disease that is unknown to humans. The features are named Symptom 1, Symptom 2, etc.

3.2 Experimental Design

This study was inspired by the work done by Hase and Bansal [10], who conducted simulation tests. Their experiment was based on the property of simulatability, described by Doshi-Velez and Kim [7] as a crucial property for interpretable models. The idea is that a model is simulatable if one can predict its behavior for new inputs. Thus, if a person can simulate the behavior of a model, it means that they understand how it works. We replicated the experiment the researchers did and adapted a few steps for our purposes. We used an online questionnaire that was sent to the respondents and they answered it voluntarily. The questionnaire included (1) an Informed Consent Form, (2) a set of profile questions and (3) the simulation questions. After collecting the answers, the data processing and analysis were done using Python programming language.

3.2.1 ML Model and XAI method. Since the focus of this study is not to evaluate a specific model's performance, and for the purposes of assessing the participants' understanding of a model's behavior, it would be interesting to have some wrongly classified instances,

we chose a Random Forest Classifier to generate the predictions. It is a model that is easy to utilize since there is a ready-made implementation in the scikit-learn Python library, and it also requires less computing power than other models, such as deep neural networks, for instance. Participants were not informed of the classifier used. The explanation method used to generate the explanations for the model was the SHAP method. SHAP is a method presented by Lundberg and Lee [19] based on Shapley values, a concept from games theory that assigns a contribution value to each game participant to get to a certain outcome. In the context of model explanation, Shapley values represent the impact that each feature has on the predictions. Considering a binary prediction task, the impact of the features can be positive (contribute to the positive class), zero (neutral), or negative (contribute to the negative class). By having the contribution of each feature on each observation of the dataset, it is possible to have local explanations (*i.e.*, to visualize how each feature contributed to a specific prediction) and global explanations (*i.e.*, to have a notion of how each feature contributes to the predictions generally).

3.2.2 Simulation. The study was conducted by using a questionnaire and following this procedure:

(1) We first present the respondents with four instances of the dataset, the real classification, and the classification given by the model so that they can observe them and try to understand how the model works, what it is doing right, and what it is doing wrong.
(2) Then we give them four new instances, without the classifications, and ask them to classify each.
(3) We give them two of the instances given in step 1 and two new ones, along with graphical explanations generated by the SHAP method.
(4) We give two of the instances given in step (2) and two new ones, along with their explanations, and ask them to classify them.
(5) We repeat step (3) and add a textual explanation to the visualization for each observation.
(6) We repeat step (4) and add a textual explanation to the visualization for each observation.

After each classification step (steps 2, 4, and 6), we ask the respondents how confident they are about their response using a 5-point Likert Scale (where 1 indicates "not confident at all" and 5 indicates "extremely confident") and also what made them choose the selected class as an open-ended question. For each step, we gave four instances having one true positive, one true negative, one false positive and one false negative according to the model's predictions. That is so the respondents could observe what the model is doing right and what it is doing wrong and then show if they can give the correct classification. Furthermore, that was a strategy so that in the classification steps, we could prevent the respondents from just guessing all the classifications and succeeding even so. The objective of this design is to mimic real situations in which the model can be erroneous, and the explanations are used to identify these mistakes.

In step 1, we give the respondents the observation without any explanation so that we can assess whether their ability to interpret the model improves when they are given the same or similar observations along with the explanation to classify. Along the steps, we always repeat two previously seen explanations and give two new ones so that we can compare the responses and observe their progress, but also not suffer from the effect of time in the respondent's learning process. Table 1 presents in which classification steps each instance is used and, consequently, how many times they were asked to be classified.

Table 1: Steps in which each instance is used and how many times they were classified

Instance	Step	Count
1	2 and 6	22
2	2 and 4	22
3	2 and 4	22
4	2 and 6	22
5	4	11
6	4	11
7	6	11
8	6	11

There are two hypotheses to be tested in this experiment:

H1 The visual explanations help people to better understand the behavior of the model.
H2 The textual explanations help people to better understand the visual explanations and improve even further the interpretability of the model in comparison with visual explanations alone.

In the experiment, we divided the respondents into three groups. Each group received one type of visualization in the steps, and they had visual explanations available for their analysis before giving the classifications. The chosen visualizations were a beeswarm chart, a bar chart, and a waterfall chart, which are generated by the Python SHAP library[20]. An example of each visualization can be seen in Figures 1, 2, and 3.

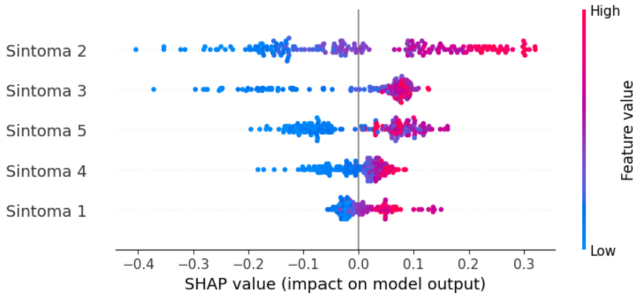

Figure 1: Beeswarm plot generated by Python SHAP library

The purpose of giving three types of visualizations for the respondents to observe was to understand whether one would be easier to interpret than the others.

One example of textual explanation is presented below. This explanation is for the graph in Figure 1:

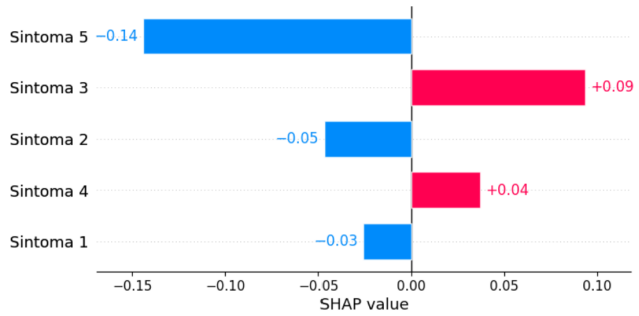

Figure 2: Bar plot generated by Python SHAP library

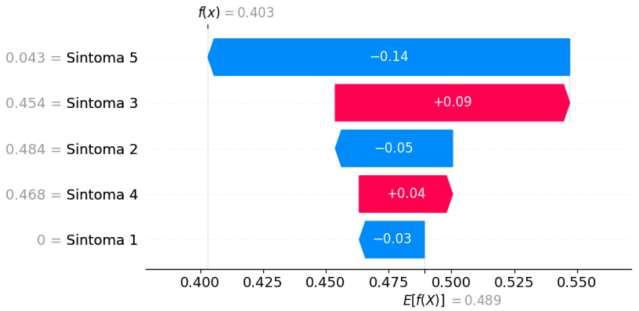

Figure 3: Waterfall plot generated by Python SHAP library

"This plot shows the features sorted by magnitude of impact on the model in general, considering their absolute mean SHAP values. Each feature has a beeswarm, and they are composed of dots that represent each observation and are distributed along the x-axis according to the observations' SHAP values. On the right side of the plot there is a vertical bar that gives the color shades of the dots that represent from high to low values of the features. Red dots represent high feature values, shades of purple represent medium values, and blue dots represent low values. That means that along with the information on how much impact each feature has on the model output, it is possible to know what ranges of feature values have positive, negative, or no impact. In this particular case, the 5 features that presented the highest impact in the model were Sintoma 2, Sintoma 3, Sintoma 5, Sintoma 4, and Sintoma 1, in that order."

3.3 Respondents' Profiles

At the beginning of the questionnaire, we asked participants to declare their free and informed consent and answer some demographic profile questions. There were 12 respondents to the questionnaire. One of them was disregarded since they did not have the required mathematical or statistical background, resulting in 11 valid responses. 6 of the respondents had a graduate degree, 4 were attending graduate courses, and 1 had completed their undergraduate studies. To evaluate their responses, having in mind their understanding of machine learning and artificial intelligence, we asked them about their level of knowledge on the matter. 7 said that they had advanced knowledge, and 4 said that they had intermediate knowledge. We also asked about their knowledge of the interpretability of machine learning models: 5 stated they had intermediate knowledge of the subject, 3 basic knowledge, and 3 advanced knowledge. Considering that the textual explanations presented to the respondents in steps 5 and 6 of the experiment were written in English, we also asked them what they consider to be their level of English. 8 were advanced, and 3 were fluent. We did not think it essential to ask about the participants' ages, but given the channels we used to recruit them, we estimate that most were 20 to 30 years old.

The list of profile questions and answer count for each option is presented in Table 2.

4 Results and Discussion

The analyses were conducted from several different perspectives. First, we analyzed the progress of the participants' understanding of the classification problem throughout the experiment as they received more information with the explanations. We also compared how each type of visualization impacted the participants' confidence and correctness. Additionally, we assessed whether the instances given for classification had differences in difficulty for classification. Finally, we evaluated how the textual explanation affected the participants' understanding of the problem.

When observing the progress of each participant throughout the questionnaire, in most of the cases, there was no improvement from the step where the respondents did not have any explanation for the instances they were asked to classify to the step where they had the visual explanation, regardless of the type of visualization. Table 3 presents the percentage of correct classifications in the first part, when the participants had to classify the instances without having any explanation; in the second part, when they had only the visual explanation; and in the third part, when they had the visual and the textual explanation. In the last part, instead of presenting some improvement, as expected, since they also had the textual explanation to help understand the visualization, there was a significant drop in the percentage of correctly classified instances. That can be either because the textual explanation generated some confusion for the respondents because the instances presented in this part were harder to classify, or because of fatigue.

Table 4 shows the overall percentages of correct and incorrect classifications per instance. This table was built with the objective of evaluating which instances are possibly easier or more difficult to classify. We found this analysis necessary when we observed that some of the instances had a tendency to be more wrongly classified than others, regardless if they had an explanation along with it or not. That difference in difficulty for classification can happen because instances that present extreme values or significant differences between variable values can be more "obvious" to classify than others that have more mild values or less variance between variables. The analysis of the impact of the explanations can be affected by this difficulty difference, since this condition can wrongly indicate that explanations are not helpful when actually the instances that do have explanations can simply be more difficult, for example. The data suggests that instances 3,7, and 8 are possibly more difficult to classify since, virtually, in every case, the instances have been wrongly classified.

Table 2: Profile questions and answer count for each response option

Question	Option	Count
What is your level of education?	Completed High School	0
	Attending Undergraduate School	0
	Completed Undergraduate School	1
	Attending Graduate School	4
	Completed Graduate School	6
Do you have a background in or are you currently enrolled in a mathematics or statistics foundational course?	Yes	11
	No	0
What do you consider to be your level of knowledge of Machine Learning and/or Artificial Intelligence?	No knowledge	0
	Basic knowledge	0
	Intermediate knowledge	4
	Advanced knowledge	7
What do you consider to be your level of knowledge of Interpretability of Machine Learning Models?	No knowledge	0
	Basic knowledge	3
	Intermediate knowledge	3
	Advanced knowledge	5
What is your level of proficiency in English?	Basic	0
	Intermediate	0
	Advanced	8
	Fluent	3

Table 3: Percentage of correct classifications in each step of the experiment

Without explanation	With visual explanation	With visual and textual explanation
50%	59%	9%

Table 4: Overall percentages of correct and incorrect classifications per instance

Instance	Correct	Incorrect
1	50%	50%
2	59%	51%
3	9%	91%
4	68%	32%
5	91%	9%
6	64%	36%
7	0%	100%
8	0%	100%

When comparing the visualizations, the one that presented the highest general success rate was the bar chart, with 40.6% of the questions correctly answered. The second one was the beeswarm plot, with 37.5% of correct answers. Last was the waterfall plot, with only 25% of the questions answered correctly (see Table 5).

Table 5: Overall performance of each visualization type

Visualization type	Correct
Bar	40.6%
Beeswarm	37.5%
Waterfall	25%

Table 6 presents the percentage of correct answers for each instance and each visualization type. With this view, we can see that instance 1, along with the previously seen instances 3, 7, and 8, is probably more difficult to predict than other instances, regardless of the visualization type.

Table 6: Percentage of correct answers for each instance and each visualization type

Instance	Beeswarm	Bar	Waterfall
1	0%	0%	0%
2	66%	75%	75%
3	33%	0%	0%
4	66%	50%	0%
5	100%	100%	75%
6	33%	100%	50%
7	0%	0%	0%
8	0%	0%	0%

In Table 7, we present the percentages of confidence levels chosen by the respondents for each visualization type. The beeswarm plot was the one that generated less confidence out of all the visualizations. The bar and the waterfall plot high percentage of high-confidence choices, but the bar plot had the highest percentage of low confidence. Thus, it is possible to conclude that out of these three visualizations, the one that generated more confidence was the waterfall plot, followed by the bar plot, and last, the beeswarm plot. We believe this is because bar and waterfall plots are much

more common than beeswarm plots. Furthermore, they represent explanations for each predicted instance, while the beeswarm plot represents a global behavior of the model. For these reasons, the beeswarm plot can be harder to interpret than the others, even for people with a strong mathematical and statistical background.

Table 7: Distribution of respondents' confidence levels for each visualization type

Visualization type	1-2	3	4-5
Bar	37%	17%	46%
Beeswarm	50%	22%	26%
Waterfall	12%	37%	50%

In contrast, it is interesting to see that the waterfall plot, which generated the highest confidence, was the one with the least correctly answered classifications. That can be a point of attention since it may indicate that this kind of visualization can possibly be misleading in some cases. In order to investigate this hypothesis further, we constructed Table 8, which has the same percentage values for each confidence level and visualization type as Table 7, but now we also have the percentages of success (green) and mistakes (red) for each visualization type for each confidence level. The waterfall plot, which we saw that might be misleading since it provided the highest confidence level but had the most mistakes, presented more cases of mistakes than successes when the respondents were most confident. Furthermore, when they said to have level 3 of confidence, which was "relatively confident", they had 100% of the instances incorrectly classified. When we look at the highest confidence levels, on the last column of the table, all of the visualization types had more incorrect than correct classifications. Additionally, by observing the table as a whole, we see that the only case in which there were more correct than incorrect answers was for the beeswarm plot when participants demonstrated low confidence. These numbers corroborate the idea that these visualizations can be misleading in some cases, and therefore their assessment before being available for use is so imperative.

Table 8: Distribution of respondents' confidence levels for each visualization type and percentages of correct (green) and incorrect (red) classifications for each visualization type for each confidence level

	Confidence level					
Visualization type	1-2		3		4-5	
Bar	37%		17%		46%	
	44%	56%	25%	75%	36%	64%
Beeswarm	50%		22%		28%	
	67%	33%	50%	50%	0%	100%
Waterfall	12%		37%		50%	
	33%	67%	0%	100%	42%	58%

After the previously described observations, we performed statistical tests to verify whether there were significant differences between the answers to the question set without the explanations, the one with the graphical explanations, and the one with the graphical and the textual explanations. The test was done by gathering the number of correctly answered questions in each question set and comparing them pairwise using the Mann Whitney U Test [9]. Table 9 presents the p-values generated by the tests that compare the question set that had the instances alone with the question sets that had the visual explanations and the one that had the visual and the textual explanations. The null hypothesis of each test is that the two compared groups are drawn from the same population. Both tests presented a p-value greater or equal to 0.05, and therefore we could not reject the null hypothesis, which means that there is no significant difference between the conditions. That indicates that there is no significant difference in the number of correctly answered classification questions between the groups that do not have explanations and those that do have.

Similarly, we performed the same statistical test, but this time comparing the groups of questions that had each of the three types of visualization. The results are presented in Table 10. Once more, the p-values were greater than 0.05, and therefore, there was no significant difference between the conditions.

The statistical tests thus showed that the availability of explanations did not make a difference in the number of instances correctly classified, nor did the type of visualization used in the graphic explanation. As we had a small number of responses and instances, the power of the tests was low, so there is a need to repeat the experiment and give more instances to classify and try to get a larger amount of responses.

The step with the textual explanation was the hardest one for most respondents, considering they made more mistakes and also presented less confidence in these questions. In these cases, we suppose that there was at least one of two possible issues: (1) the presented instances might have been too hard to interpret, or (2) the textual explanation made the respondents more confused instead of helping them to interpret the visualization. Some participants who claimed to have advanced knowledge of Machine Learning and model interpretability seemed to have a good understanding of the visual explanation in their answers to the open-ended questions, yet still got most of the answers wrong. Additionally, one of the participants said that they did not understand the visualization when they did not have the textual explanation and seemed to have understood it better when they had it in hand. However, they got more wrong answers in the last case. All of those observations, along with the analysis presented in Table 4 that showed two of the instances used in this step as some of the hardest ones to predict, made us believe that the instances selected for that part of the experiment were indeed too hard to predict.

We observed another issue with the understanding of what was being asked in the questionnaire since some respondents said that they were not sure if they should answer what they thought to be the right classification or the one that they thought the model would choose. The amount of instances provided for observation seemed to have been an issue as well, some of the respondents said that they needed more instances to understand the problem better.

With the results obtained, we understood that the visual explanations given by the Python SHAP library can be hard to understand even for people who have a mathematical and/or statistical background and claim to have a good understanding of machine learning

Table 9: Results of the Mann Whitney U Test of responses without and with explanations

Pair of question set	p-value
Instances without explanation vs Instances with visual explanations	0.77
Instances without explanation vs Instances with visual and textual explanations	0.05

Table 10: Results of the Mann Whitney U Test of responses using different visualizations as explanations

Pair of visualization type group	p-value
Beeswarm plot vs Bar plot	0.74
Beeswarm plot vs Waterfall plot	0.73
Bar plot vs Waterfall plot	0.49

and model interpretability. Therefore we had to reject **H1**, that said that visual explanations would help users to better understand the behavior of a model. It is possible that visualizations are efficient in other cases, but we through this experiment we understood that it is not guaranteed that any visualization will achieve this objective. Furthermore, since the results on the last part of the experiment also did not show improvement in the classifications of the participants, we had to reject **H2** as well. Again, it is likely that other textual explanations would be more efficient that the one we gave to the participants, but even having a textual explanation it is not possible to guarantee that there will be an improvement of the understanding of the model's behavior.

In the study conducted by Hase and Bansal [10] they evaluated a few different explanation methods, and the only one which they found definitive improvement in the understanding of the participants of the model's behavior was for the LIME method. All the other four evaluated methods were not able to confer this improvement. This shows that our study is in accordance with the findings of other existing study, that also emphasized that there is a need for methods that are efficient in evaluating explanations.

5 Study Limitations

The study presents some limitations that are worth discussing. The number of respondents (12) was low, which means the power of any statistical tests is also low, making it more difficult to find statistically significant differences if they exist. In future studies, we intend to use additional channels of communication to publicize it and leave the questionnaire open for longer, in an attempt to gather data from a larger pool of respondents.

We found a relevant internal threat to validity. It is possible that incorrect classifications were not due to the condition itself (type of visualization and presence or absence of a textual explanation) but to the level of difficulty of the instances presented in the different conditions. Care must be taken to better balance the degree of difficulty of the instances across all conditions to avoid this kind of interference in the results.

In the study reported here, presenting a textual explanation alongside the visualizations hindered respondents' interpretation. However, because we wanted to keep the questionnaire sufficiently short to avoid respondents' fatigue, we generated a single style of explanatory text, so we cannot conclude that other kinds of textual explanations would yield similar results. As such, future studies should investigate how different kinds of explanatory text influence the respondents' interpretation.

The study was conducted by questionnaire, which limits the collection of rich, in-depth qualitative data. To complement the study reported here, we are currently designing an observational and interview study to investigate the reasoning underlying the interpretation and construction of visualizations to support XAI.

6 Conclusion

As we have observed, explanations presented by existing explanation methods can be hard to understand even for people in the mathematical and statistical fields, so it is likely even more difficult for lay users. In scenarios where machine learning models are being used for critical tasks that require a high level of trust and transparency, those methods can still be insufficient for clarifying how a model reached a certain decision and ensuring it can be trusted. These methods have started paving a path to improving machine learning models' interpretability, but there is still a long way to go in order to actually achieve full transparency, trust, and accountability.

The study we conducted suggests that the explanations generated by the SHAP library may not be sufficiently helpful for improving the model's interpretability in some cases. That can be because the visualizations generated by the method are hard to interpret, thus inefficient in helping users understand the model's behavior. Furthermore, we found that some visualizations can even be misleading since they confer more confidence to the user, even though they are mistaken in their classification.

There were a few issues with the experiment that made it not fully conclusive. Future work should reproduce the experiment, giving more instances as examples to the respondents so that they can get a wider view of the problem and also more instances to classify so that we get more data points to analyze. Furthermore, there should be a way to guarantee that the selected instances have similar difficulty levels so that the complexity factor does not impact the experiment and that the results can be more reliable.

Despite the issues observed in the experiment, it made us confident that XAI and VA researchers should certainly work collaboratively to develop visualizations that are more efficient in helping the user interpret ML models, regardless of whether they are end users, ML specialists, or developers. Additionally, the proposed visualizations should always be evaluated with various audiences since each could have a different understanding of the explanations, depending on their background and domain knowledge. Future work could develop new visualizations with the assistance of VA researchers and then test them with target users with different backgrounds and diverse domains to assess their effectiveness.

Acknowledgments

The authors thank the volunteer study participants for making this work possible. They also thank CNPq (process # 313049/2021-1) and CAPES for the financial support of this work.

References

[1] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. 2018. Sanity Checks for Saliency Maps. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html

[2] Sajid Ali, Tamer Abuhmed, Shaker El-Sappagh, Khan Muhammad, Jose M. Alonso-Moral, Roberto Confalonieri, Riccardo Guidotti, Javier Del Ser, Natalia Díaz-Rodríguez, and Francisco Herrera. 2023. Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence. *Information Fusion* 99 (Nov. 2023), 101805. https://doi.org/10.1016/j.inffus.2023.101805

[3] Gulsum Alicioglu and Bo Sun. 2022. A survey of visual analytics for Explainable Artificial Intelligence methods. *Computers & Graphics* 102 (Feb. 2022), 502–520. https://doi.org/10.1016/j.cag.2021.09.002

[4] Or Biran and Courtenay Cotton. 2017. Explanation and justification in machine learning: A survey. In *IJCAI-17 workshop on explainable AI (XAI)*, Vol. 8. 8–13.

[5] A. Chatzimparmpas, R. M. Martins, I. Jusufi, K. Kucher, F. Rossi, and A. Kerren. 2020. The State of the Art in Enhancing Trust in Machine Learning Models with the Use of Visualizations. *Computer Graphics Forum* 39, 3 (2020), 713–756. https://doi.org/10.1111/cgf.14034 _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.14034.

[6] Michael Correll. 2019. Ethical Dimensions of Visualization Research. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–13. https://doi.org/10.1145/3290605.3300418 arXiv:1811.07271 [cs].

[7] Finale Doshi-Velez and Been Kim. 2017. Towards A Rigorous Science of Interpretable Machine Learning. http://arxiv.org/abs/1702.08608 arXiv:1702.08608 [cs, stat].

[8] David Gunning and David W. Aha. 2019. DARPA's Explainable Artificial Intelligence Program. *AI Magazine* 40, 2 (2019), 44–58. https://doi.org/10.1609/aimag.v40i2.2850 arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1609/aimag.v40i2.2850

[9] A. Hart. 2001. Mann-Whitney test is not just a test of medians: differences in spread can be important. *BMJ* 323, 7309 (Aug. 2001), 391–393. https://doi.org/10.1136/bmj.323.7309.391

[10] Peter Hase and Mohit Bansal. 2020. Evaluating Explainable AI: Which Algorithmic Explanations Help Users Predict Model Behavior? http://arxiv.org/abs/2005.01831 arXiv:2005.01831 [cs].

[11] Robert R Hoffman, Shane T Mueller, Gary Klein, and Jordan Litman. 2018. Metrics for Explainable AI: Challenges and Prospects. (2018).

[12] Been Kim. 2015. Interactive and Interpretable Machine Learning Models for Human Machine Collaboration. (2015).

[13] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. 2018. Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV). http://arxiv.org/abs/1711.11279 arXiv:1711.11279 [stat].

[14] Himabindu Lakkaraju, Stephen H. Bach, and Jure Leskovec. 2016. Interpretable Decision Sets: A Joint Framework for Description and Prediction. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, San Francisco California USA, 1675–1684. https://doi.org/10.1145/2939672.2939874

[15] Himabindu Lakkaraju, Ece Kamar, Rich Caruana, and Jure Leskovec. 2019. Faithful and Customizable Explanations of Black Box Models. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*. ACM, Honolulu HI USA, 131–138. https://doi.org/10.1145/3306618.3314229

[16] Yiran Li, Takanori Fujiwara, Yong K. Choi, Katherine K. Kim, and Kwan-Liu Ma. 2020. A visual analytics system for multi-model comparison on clinical data predictions. *Visual Informatics* 4, 2 (June 2020), 122–131. https://doi.org/10.1016/j.visinf.2020.04.005

[17] Zachary C. Lipton. 2017. The Mythos of Model Interpretability. http://arxiv.org/abs/1606.03490 arXiv:1606.03490 [cs, stat].

[18] Yin Lou, Rich Caruana, Johannes Gehrke, and Giles Hooker. 2013. Accurate intelligible models with pairwise interactions. In *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*. ACM, Chicago Illinois USA, 623–631. https://doi.org/10.1145/2487575.2487579

[19] Scott M Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[20] Scott M Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems 30*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett (Eds.). Curran Associates, Inc., 4765–4774. http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf

[21] Tim Miller. 2019. Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence* 267 (Feb. 2019), 1–38. https://doi.org/10.1016/j.artint.2018.07.007

[22] Brent Mittelstadt, Chris Russell, and Sandra Wachter. 2019. Explaining Explanations in AI. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. Association for Computing Machinery, New York, NY, USA, 279–288. https://doi.org/10.1145/3287560.3287574

[23] Brett Poulin, Roman Eisner, Duane Szafron, Paul Lu, Russ Greiner, D S Wishart, Alona Fyshe, Brandon Pearcy, Cam MacDonell, and John Anvik. 2006. Visual Explanation of Evidence in Additive Classifiers. (2006).

[24] Oliver Radley-Gardner, Hugh Beale, and Reinhard Zimmermann (Eds.). 2016. *Fundamental Texts On European Private Law*. Hart Publishing. https://doi.org/10.5040/9781782258674

[25] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. http://arxiv.org/abs/1602.04938 arXiv:1602.04938 [cs, stat].

[26] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2018. Anchors: High-Precision Model-Agnostic Explanations. *Proceedings of the AAAI Conference on Artificial Intelligence* 32, 1 (April 2018). https://doi.org/10.1609/aaai.v32i1.11491

[27] Greg Ridgeway, David Madigan, Thomas Richardson, and John O'Kane. 1998. Interpretable Boosted Naïve Bayes Classification. (1998).

[28] Dylan Slack, Sophie Hilgard, Emily Jia, Sameer Singh, and Himabindu Lakkaraju. 2020. Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods. http://arxiv.org/abs/1911.02508 arXiv:1911.02508 [cs, stat].

[29] J W Smith, J E Everhart, W C Dickson, W C Knowler, and R S Johannes. 1988. Using the ADAP Learning Algorithm to Forecast the Onset of Diabetes Mellitus. *Proc Annu Symp Comput Appl Med Care* (1988), 261–265.

[30] Thilo Spinner, Udo Schlegel, Hanna Schafer, and Mennatallah El-Assady. 2019. explAIner: A Visual Analytics Framework for Interactive and Explainable Machine Learning. *IEEE Transactions on Visualization and Computer Graphics* (2019), 1–1. https://doi.org/10.1109/TVCG.2019.2934629

[31] Paolo Tamagnini, Josua Krause, Aritra Dasgupta, and Enrico Bertini. 2017. Interpreting Black-Box Classifiers Using Instance-Level Visual Explanations. In *Proceedings of the 2nd Workshop on Human-In-the-Loop Data Analytics*. ACM, Chicago IL USA, 1–6. https://doi.org/10.1145/3077257.3077260

[32] Claire Woodcock, Brent Mittelstadt, Dan Busbridge, and Grant Blank. 2021. The Impact of Explanations on Layperson Trust in Artificial Intelligence–Driven Symptom Checker Apps: Experimental Study. *Journal of Medical Internet Research* 23, 11 (Nov. 2021), e29386. https://doi.org/10.2196/29386

Received 19 May 2024; revised 19 July 2024; accepted